



Building a Cyberattack Classifier

Mohammed Omar Mkhallati

AI for Cybersecurity

https://colab.research.google.com/github/omarmkhallati/AI_For_Cybersecurity-Mkhallati/blob/main/Mohammed_Omar_Mkhallati-CyberAttack_Classifier.ipynb

Submitted Monday, September 14, 2026

Part A – Dataset Understanding

I will be using the CEAS_08 dataset since I found it to be the clearest. My model will have 3 input features. Firstly, there is the content of the email which includes the subject and the body. Secondly, there is the sender domain, since some domains are repeated abusers and always send phishing emails so we might see a pattern. Finally, my third feature is the presence of URLs in the email. The target labels are a malicious/phishing email (1) or a safe email (0). This is a binary classification problem to figure out whether an email is malicious or not. It is a supervised learning problem since we have labelled data with the value of it being 0 or 1 to train on. An unsupervised learning problem would have unlabeled data, and we would not have values for malicious or safe.

Part B – Model Selection

The most suitable algorithm is logistic regression. This fits the dataset because once the email content is converted into word-based features, the data becomes high dimensional and sparse, meaning most values are zero for any single email, and logistic regression handles this kind of data efficiently without overfitting the way more complex models might. It also works naturally alongside the sender domain and URL presence features once everything is converted into numbers. Another reason it fits is that it outputs a probability rather than a strict yes or no, so we can adjust how cautious the model is depending on how strict we want to be about flagging and phishing emails. It also stays interpretable, since each feature gets a weight showing whether it pushes a prediction toward phishing or safe, which is useful for explaining why an email was flagged.

Part C – Cybersecurity Analysis

The model classifies phishing emails, a form of social engineering where an attacker pretends to be a trusted sender to trick someone into revealing credentials, clicking a malicious link, or transferring money. Because the model factors in the email's content, the sender's domain, and whether a link is present, it targets the combination of signals that make phishing recognizable, not just suspicious wording on its own. Classification matters in cybersecurity because it lets an organization filter or flag suspicious emails automatically, before an employee ever opens them. Given how much email a company receives daily, manually reviewing every message is not realistic, so a model like this acts as a first line of defense, catching most phishing attempts before they reach an inbox and reducing the chance someone falls for one. If the model misclassifies, the consequences differ depending on the type of error. A false negative, where a phishing email is labeled legitimate, lets the attack reach the user undetected, which can lead to stolen credentials, financial loss, or a malware infection. A false positive, where a legitimate email is flagged as phishing, can block or delay real communication and, if it happens often enough, teach employees to distrust or ignore the filter altogether, which undermines the point of having it in the first place.